

Writing Decision - Making Statements

- Decision - making statements evaluate conditions and execute statements based on that evaluation
- VB .NET includes:
 - If and Select Case statements

➤ If statement

- Evaluates an expression.
- Executes one or more statements if expression is true.
- Can execute another statement or group of statements if expression is false

➤ Select Case statement

- Evaluates a variable for multiple values.
- Executes a statement or group of statements, depending on contents of the variable being evaluated.

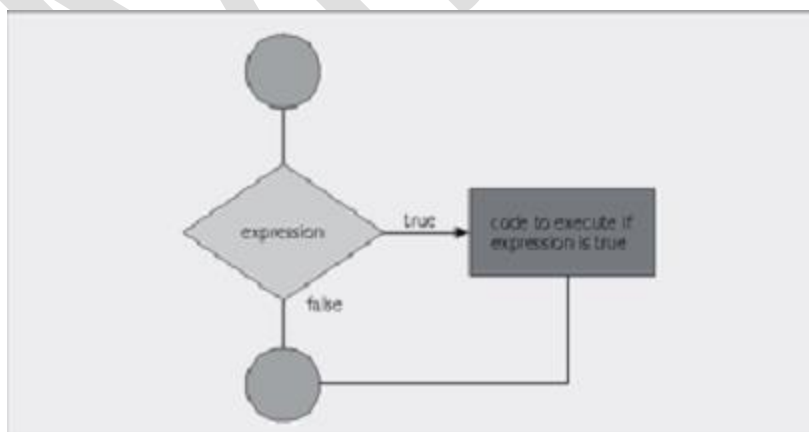
Writing If Statements

- ✚ VB .NET If statement interrogates a logical expression that evaluates to true or false.
- ✚ An expression often compares two values using logical operators.

VB.Net Logical Operators

Operator	Description
=	Equal to
>	Greater Than
>=	Greater Than or Equal to
<	Less Than
<=	Less Than or Equal to
<>	No Equal to

- VB.NET If statement has two forms:
Simple If and If-Else
- **Simple If**
 - Evaluates an expression.
 - Executes one or more statements if expression is True.



Simple If Statement Logic

The following syntax for If Then Else Statement

IF Condition Then

Statement(s)

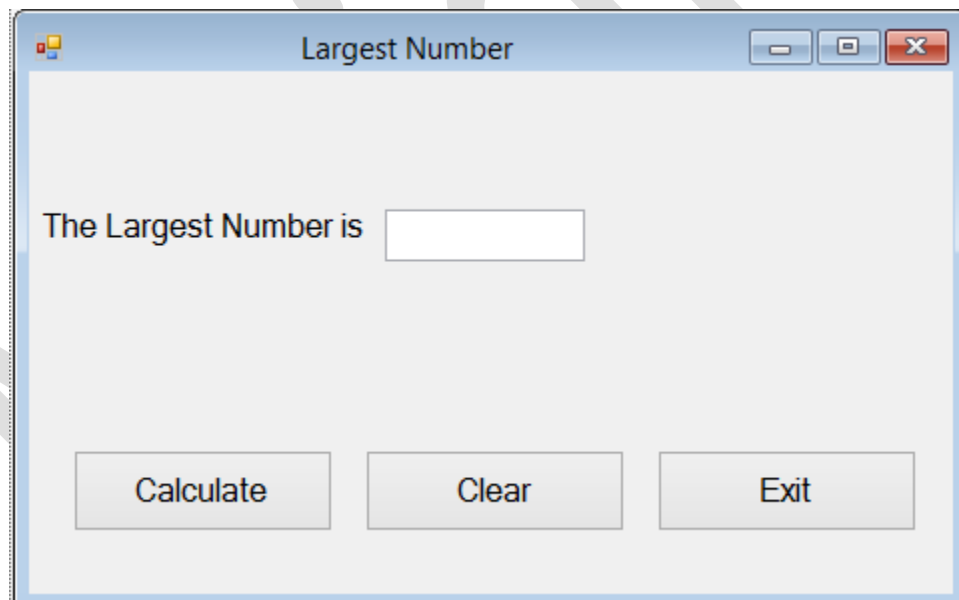
Else

Statement(s)

End IF

Example:

Design the following form and write a program in VB.Net to read two integer numbers then find the largest number between them?



The screenshot shows a Windows application window with the title bar 'Largest Number'. Inside the window, there is a label 'The Largest Number is' followed by a text input field. Below the input field, there are three buttons arranged horizontally: 'Calculate', 'Clear', and 'Exit'.

The Solution is:

To design the above form we implement the following set properties of the objects:

Object	Property	Setting
Button1	Text	Calculate
Button2	Text	Clear
Button3	Text	Exit
Label1	Text	The Largest Number is
Form1	Text	Largest Number
Textbox1		

Writing the Code

Use the Code Editor

1. Double-click the Calculate Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim X, Y As Integer
    X = InputBox("Enter The First Number")
    Y = InputBox("Enter The Second Number")
    If X >= Y Then
        TextBox1.Text = X
    Else
        TextBox1.Text = Y
    End If
End Sub
```

2. Double-click the Clear Button2 on the form and write the following code inside it:

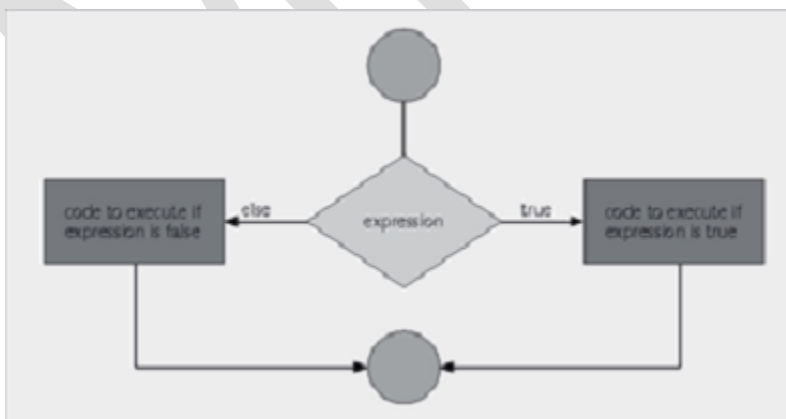
```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click  
    TextBox1.Clear()  
End Sub
```

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click  
    End  
End Sub
```

▪ If-Else.

- Evaluates an expression
- Executes one or more statements if expression is true.
- Executes a second statement or set of statements if expression is false.



If-Else Statement Logic

The following syntax for If Then ElseIf Statement:

IF Condition1 Then

Statement1(s)

ElseIf Condition2 Then

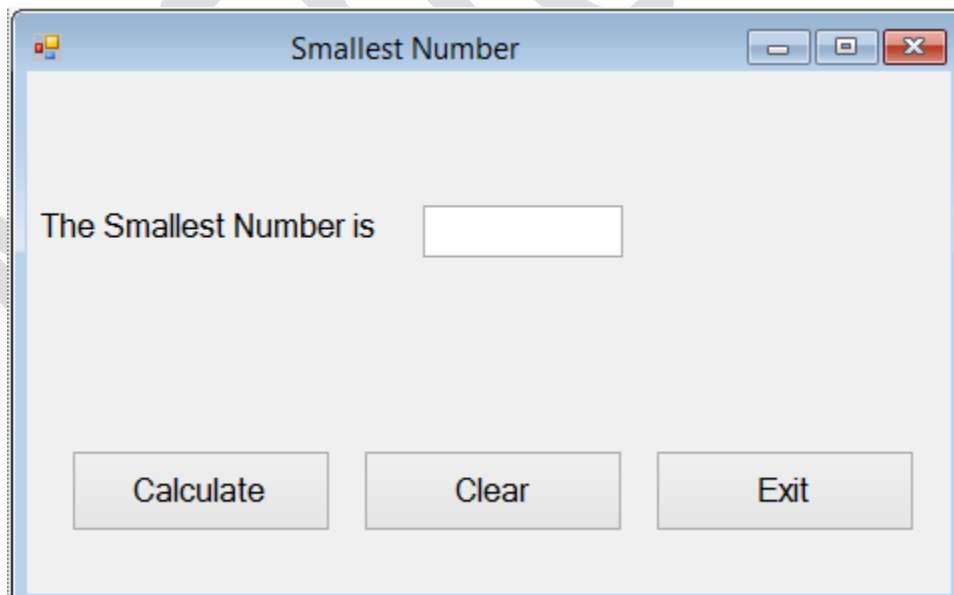
Statement2(s)

Else Statement

End IF

Example:

Design the following form and write a program in VB.Net to read three integer numbers then find the smallest number between them?



Smallest Number

The Smallest Number is

Calculate Clear Exit

The Solution is:

To design the above form we implement the following set properties of the objects:

Object	Property	Setting
Button1	Text	Calculate
Button2	Text	Clear
Button3	Text	Exit
Label1	Text	The Smallest Number is
Form1	Text	Smallest Number
Textbox1		

Writing the Code

Use the Code Editor

1. Double-click the Calculate Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim X, Y, Z As Integer
    X = InputBox("Enter The First Number")
    Y = InputBox("Enter The Second Number")
    Z = InputBox("Enter The Third Number")
    If X < Y And X < Z Then
        TextBox1.Text = X
    ElseIf Y < X And Y < Z Then
        TextBox1.Text = Y
    ElseIf Z < X And Z < Y Then
        TextBox1.Text = Z
    Else
        MessageBox.Show("The Numbers are Equal")
    End If
End Sub
```

End If

End Sub

2. Double-click the Clear Button2 on the form and write the following code inside it:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click  
    TextBox1.Clear()
```

End Sub

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
```

End

End Sub

Writing Select Case Statements

- Select Case statement
 - Acts like a multiple-way If statement.
 - Transfers control to one of several statements, depending on the value of an expression.

The following syntax for Select Case Statement:

Select Case Expression

Case Value list

Statement(s)

Case Else

Statement(s)

End Select

Example:

Write a program in VB.Net to read number in the range of 1 to 7, a message box display the weekday depending on the number entered. If the number entered is not in the range 1 to 7 a message box appears The Number is Out of the Range?

Writing the Code

Use the Code Editor

- Double-click the Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
```

```
    Dim X As Integer
```

```
    X = InputBox("Enter The Number Between 1 to 7 ")
```

```
    Select Case X
```

```
        Case 1
```

```
            MessageBox.Show("Sunday")
```

```
        Case 2
```

```
            MessageBox.Show("Monday")
```

```
        Case 3
```

```
            MessageBox.Show("Tuesday")
```

```
        Case 4
```

```
            MessageBox.Show("Wednesday")
```

```
        Case 5
```

```
            MessageBox.Show("Thursday")
```

```
        Case 6
```

```
            MessageBox.Show("Friday")
```

```
        Case 7
```

```
            MessageBox.Show("Saturday")
```

```
        Case Else
```

```
            MessageBox.Show("The Number is out of Range")
```

```
    End Select
```

```
End Sub
```